Modelling the electromagnetic response of a sphere located in a layered earth

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Abstract

Estimating the electromagnetic response of a conductive sphere in a layered earth is of great interest in terms of both modelling and interpretating data acquired via geophysical electromagnetic methods where the target is at some distance from the source and receiver. This is particularly the case when using Airborne Electromagnetic Method (AEM) where the source and receiver are located at some height above the subsurface. This problem can be solved by utilizing field expansions representing derivatives of cylindrical functions, which describe the fields propagating in the layered earth, and spherical functions, which describe the fields reflected by the sphere. Furthermore, these representations allow the development of relationships between cylindrical and spherical functions. These functions and subsequent relationships have been used to develop an algorithm for estimating the electromagnetic response of a conductive sphere in an isotropic layered earth. Software based on this algorithm have been tested on both synthetic and field data.

The field data presented were collected with the AEM AirTEM system over the Reid-Mahaffy test site, Ontario, Canada. Results from these tests prove the importance and utility of integrating the sphere in a layered earth model in the AEM interpretation toolbox.

Introduction

The functional form of analytic solutions is usually dictated by the geometry of the problem. Fourier functions are generally preferred to generate solutions for Cartesian system coordinates, while Bessel functions are selected for cylindrical coordinates, and spherical functions are more adequate for spherical geometry. Unfortunately, the earth subsurface is not so simple and in practice a specific problem may involve a variety of complex geometries. A typical case in electromagnetic exploration is a conductive sphere in a layered earth. While there are several solutions for the response of a conducting sphere (March 1953; Wait 1953; Grant and West 1965; Lodha and West 1976; Ward and Hohmann 1987), the sphere must be located in a uniform space, which in practice is a whole space either resistive or conductive (Singh 1973). Lee has proposed in successive papers (Lee 1975; Lee 1980; Lee 1983) numerical solutions for a sphere in a layered earth based on integral equations. An alternative has been the approximation proposed by Desmarais and Smith (2016) for a sphere located below a conductive overburden. During the same period, there have been numerous developments to solve the response of a layered earth and its application to numerical methods simulating the response of an anomalous body located in a layered earth (Hohmann 1987, Raiche et al. 2007). Noteworthy inversion methods include magnetic and electric dipoles in a layered earth (Sattel and Reid 2006), plates in a 1D earth (Macnae 2015), and resistive limit inversion (Shaa 2010, Shaa and Fullagar 2010).

The seminal papers of Gumerov and Duraiswami (2003, 2007) have led to remarkable advancements in spherical field computation. In summary, by developing new relations describing derivative of sums of spherical functions, they propose a new approach to EM modelling where the goal is not to analytically derive EM fields from potentials, but to solve the fields' coefficients directly from the potential spherical coefficient derivatives. We demonstrate that this approach

can be extended to cylindrical functions as well, from which we develop a relationship between cylindrical and spherical functions. This allows the representation of the spherical fields originating from a sphere target as cylindrical fields which are evaluated with already developed Hankel transform digital filters (Christensen 1990). Using this material, we propose an algorithm for modelling the AEM response of a sphere in a layered earth. This algorithm has been implemented in a computer program dedicated to AEM modelling and interpretation, where the sphere in a layered earth model is particularly suited (Vallée 2015). We conclude by showing synthetic examples of the sphere response in various host rock environments, either a conductive half-space, a conductive thin-sheet or a layered earth, and a field example from the Reid-Mahaffy test site, Ontario, Canada.

Theory

Our solution for the response of a sphere in a layered earth follows the development of links between field descriptions in Cartesian or cylindrical coordinates and field description in spherical coordinates. In this process, we also use new developments expressing derivatives as expansions of cylindrical or spherical functions. Following Ward and Hohmann (1987, p. 145-147), we derive the electric and magnetic fields **E** and **H** from Schelkunoff (1943) potentials attached to the sources. For electric sources

$$\mathbf{E}_e = -\hat{z}\mathbf{A} + \frac{1}{\hat{y}}\nabla(\nabla \cdot \mathbf{A}) \tag{1}$$

$$\mathbf{H}_e = \nabla \times \mathbf{A} \tag{2}$$

and for magnetic sources

$$\mathbf{E}_m = -\nabla \times \mathbf{F} \tag{3}$$

$$\mathbf{H}_m = -\hat{y}\mathbf{F} + \frac{1}{\hat{z}}\nabla(\nabla \cdot \mathbf{F}) \tag{4}$$

where $\hat{y} = \sigma + i\omega\epsilon$, $\hat{z} = i\omega\mu$, $k^2 = -\hat{y}\hat{z}$, σ is the conductivity, ϵ is permittivity, μ is the magnetic permeability, ω is the angular frequency, and **A** and **F** are electric and magnetic source Schelkunoff potentials (electric source potentials are required to describe the sphere response). We assume $e^{i\omega t}$

dependency. In a horizontal layered earth made of homogeneous layers, Ward and Hohmann (1987, p. 147-148) show that all electromagnetic fields can be derived from two independent vertical A_z and F_z potentials, called transverse magnetic (TM) and transverse electric (TE).

Furthermore, wave propagation from a point source in a homogeneous medium can be developed in cylindrical coordinates using the Sommerfeld (1909) integral which describes a point source field as a Hankel transform

$$\frac{e^{-ikr}}{r} = \int_0^\infty \frac{\lambda}{u} e^{-u|z|} J_0(\lambda\rho) d\lambda \tag{5}$$

where $\rho = x^2 + y^2$, $r^2 = \rho^2 + z^2$, $u^2 = \lambda^2 - k^2$, and J_0 is the Bessel function of order zero. A general solution for the field of magnetic dipoles in a layered earth can be developed from derivatives of equation (5) as shown by Ward and Hohmann (1987, p. 203-252).

Similarly, fields incident and reflected by a sphere can be derived from two independent potentials transverse to the radius of the sphere. Solutions of E and H are expressed as sums of spherical functions propagating inward (singular, $S_n^m(\mathbf{r})$) and outward (regular, $R_n^m(\mathbf{r})$) (see complete definition in Appendix A). In practice, field computation requires the development of derivatives of potentials which can be a laborious process as demonstrated by Ward and Hohmann (1987, p. 296-304). Fortunately, it is possible to use the alternative solution developed by Gumerov and Duraiswami (2003, 2007). As detailed in Appendix A, from the derivatives of spherical functions, it is possible to develop derivatives of spherical fields in the following way. We start from

$$\sum_{n=0}^{\infty} \sum_{m=-n}^{n} \hat{q}_n^m T_n^m(\mathbf{r}) = \frac{\partial}{\partial w} \sum_{n'=0}^{\infty} \sum_{m'=-n'}^{n'} q_{n'}^{m'} T_{n'}^{m'}(\mathbf{r}) \tag{6}$$

where $T = S, R$ denotes an arbitrary inward or outward propagating spherical function, q_n^m and $\hat{q}_n^m$ are spherical coefficients, and w is either $x + iy, x - iy$ or z. n and m are the degree and order of the spherical function and the range of m values increases with n. $\hat{q}_n^m$ can be developed as a

sparse matrix operation which expression is

$$\hat{q}_n^m = \sum_{n'=0}^{\infty} \sum_{m'=-n'}^{n'} (D_w)_{nn'}^{mm'} q_{n'}^{m'} \tag{7}$$

(see Appendix A for a complete description of matrix D_w).

The same approach can be followed to develop cylindrical function derivatives. If we represent a cylindrical function as

$$I_m(\rho, \phi, z) = \int_0^{\infty} \lambda J_{|m|}(\lambda\rho) e^{-u|z|} e^{im\phi} d\lambda = \int_0^{\infty} \lambda G_m(\rho, \lambda, \phi, z) d\lambda \tag{8}$$

Derivative cylindrical coefficient expansions can be developed in the same way as

$$\sum_{m=0}^{\infty} \hat{\chi}_m G_m(\rho, \lambda, \phi, z) = \frac{\partial}{\partial w} \sum_{m'=0}^{\infty} \chi_{m'} G_{m'}(\rho, \lambda, \phi, z) \tag{9}$$

and the derivative coefficients can be estimated from cylindrical expansion (see Appendix B)

$$\hat{\chi}_m = \sum_{m'=-\infty}^{\infty} (D_w)_{mm'} \chi_{m'} \tag{10}$$

where χ_m are cylindrical coefficients. The estimation of Cartesian coordinate derivatives of cylindrical fields as expansions of cylindrical fields simplifies the derivation of analytic solutions. Furthermore, using derivative relations, we can express singular spherical functions as cylindrical expansions where

$$S_n^m(\mathbf{r}) = \int_0^{\infty} \sum_{m'=-\infty}^{\infty} \zeta_{nm'}^m G_{m'} \lambda d\lambda \tag{11}$$

where $\zeta_{nm'}^m$ is developed in Appendix B. This approach allows the computation of the fields reflected by a sphere in a layered earth. As we cannot develop the spherical functions from the cylindrical functions with a similar relation, the incident spherical field is estimated using spherical harmonic analysis (Adams and Swarztrauber, 1998).

The field originating from the sphere is reflected by the layer boundaries. This reflection is a function of the spherical harmonics and can be estimated if we represent spherical potentials by vectors such as

$$v = \begin{bmatrix} (\psi_a)_1^{-1} \\ (\psi_a)_1^{0} \\ (\psi_a)_1^{1} \\ (\psi_a)_2^{-2} \\ (\psi_a)_2^{-1} \\ (\psi_a)_2^{0} \\ ... \\ (\psi_f)_1^{-1} \\ ... \end{bmatrix} \tag{12}$$

where $(\psi_a)_n^m$ and $(\psi_f)_n^m$ are the coefficients of the radial transverse magnetic and electric potentials respectively. As the incident field on the sphere is a sum on the field from the EM source and the field reflected by the layered earth, we can write

$$v_i = v_s + v_r \tag{13}$$

where:

v_i: potential vector incident on the sphere,

v_s: potential vector from the EM source,

v_r: reflected potential vector.

Introducing the layered earth reflection as a matrix operation,

$$v_r = Cv_i = v_i - v_s \tag{14}$$

where C is a matrix where each column is the response of one line on the incident potential vector. This matrix is only a function of the electrical parameters of the sphere and the layered earth and

can be evaluated independently from the transmitter and receiver locations. Furthermore, we can write

$$v_i - Cv_i = v_s \tag{15}$$

$$v_i = (I - C)^{-1} v_s \tag{16}$$

This last equation relates sphere incident fields to source fields and allows the correction of the source primary field for the layered earth reflection.

Algorithm

Following these developments, we propose the following algorithm for estimating the electromagnetic response of a conducting sphere in a layered earth, which is illustrated by the flow chart in Figure 1. Numbers in the discussion refer to the steps of the flow chart.

Initial parameters include the locations of the transmitter and receiver, frequency or time, AEM system parameters, layered earth (thicknesses and conductivities) and sphere parameters (radius, conductivity, depth) (steps 1, 2, and 5). If it is a time domain survey, computations are done in the frequency domain and transformed to the time domain following Raiche (1998). For each frequency, we compute the sphere reflection coefficients (Ward and Hohmann, p. 301) (step 3) and the layered earth reflection correction described by equation (16) (step 4), as these computations are independent from the transmitter or receiver locations.

Then, for each survey location (step 5) we compute the primary field on the surface of the sphere following layered earth propagation equations as described by Vallée (2015), which have been simplified using equation (10) (step 7). We then estimate the primary spherical potential using spherical harmonic analysis (step 8), which involves: estimating the spherical coefficients of the two radial fields E and H, and converting the values to potentials following the relationships between radial fields and potentials described by Gumerov and Duraiswami (2007). After correcting for the layered earth reflection which as described by equation (16) in step 8, we develop the

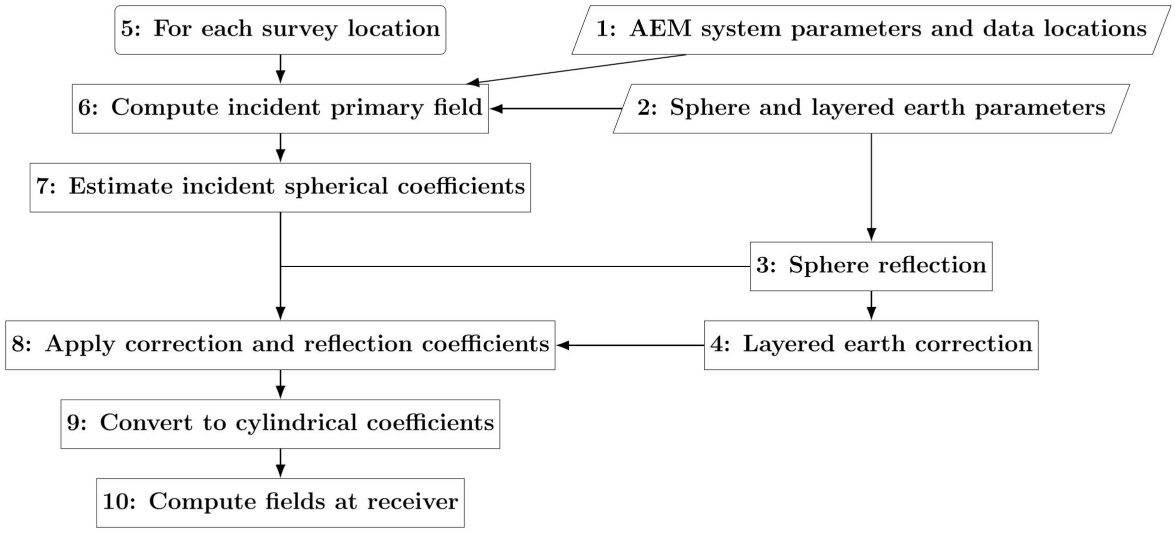

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Figure 1: Flow chart depicting the modelling algorithm. Each number refers to a step in the process which is then explained further in the following text.

spherical fields as cylindrical expansions using equation (11) (step 9), allowing the computation of fields above the earth's surface (step 10).

This model has been linked to a SVD non-linear least square inversion using a modified Gauss-Newton-Marquardt method to invert for the parameters of the sphere and/or layered earth (Raiche et al. 2006), which permits estimation of layered earth or sphere parameters such as depth, radius and conductivity. Hankel transform estimations are done using fast Hankel transform filters (Christensen 1990). A computer program has been developed following this algorithm. A graphical user interface was developed in Python, which was interfaced to a FORTRAN program for computing a model or an inversion.

Synthetic results

The accuracy of the proposed method is presented as a function of various parameters that we explore in this section before presenting some results of synthetic modelling. We first compare the results of the sphere in a conductive half-space with the free air model of Grant and West (1965). We then examine the layered earth response in time domain for two distinct programs. Finally, we present some nomograms for various scenarios including: sphere in a half-space, sphere below a thin sheet, and sphere in a layered earth.

For a comparison of responses for a sphere in free space and half-space, respectively, we limit our approach to a simple frequency-domain AEM system with the transmitter and receiver axis located at the same elevation with both a vertical axis (co-planar), spaced by 10 m. In this scenario, the system is 40 m above the ground and the sphere has a radius of 50 m and is centered 100 m below the surface with a conductivity of 1 S/m. The half-space conductivity is 0.1 mS/m. Figure 2 presents the profile response above the center of the sphere. The response difference between the two models is less than 1 % of the maximum anomaly. As the response peaks over the center of the sphere, we investigate the maximum response variation with frequency. This is presented in Figure 3 for frequencies ranging between 10 Hz and 100 kHz. Below 1 kHz, the impact of half-space is negligible, as could be already observed in Figure 2. The half-space response becomes more important above 1 kHz, particularly the quadrature response. Above 10 kHz, the sphere in the half-space response starts diverging from the free space sphere response.

To explore the impact of the layered response computation and the frequency to time transformation on the model computation, we use the parameters of the AirTEM system (Balch et al. 2016). This employs a horizontal loop transmitter with a diameter of 4.25 m and a vertical magnetic coil receiver (dB/dt) located on the side of the transmitter loop, sitting approximately 40 m above the ground. The transmitter waveform is a 1.85 ms bipolar triangular pulse with a repetition rate of 90 Hz. The time derivative of the magnetic field is sampled with 40 off-time channels between 0.02 and 3.3 ms. In Figure 4, we compared the half-space response for two

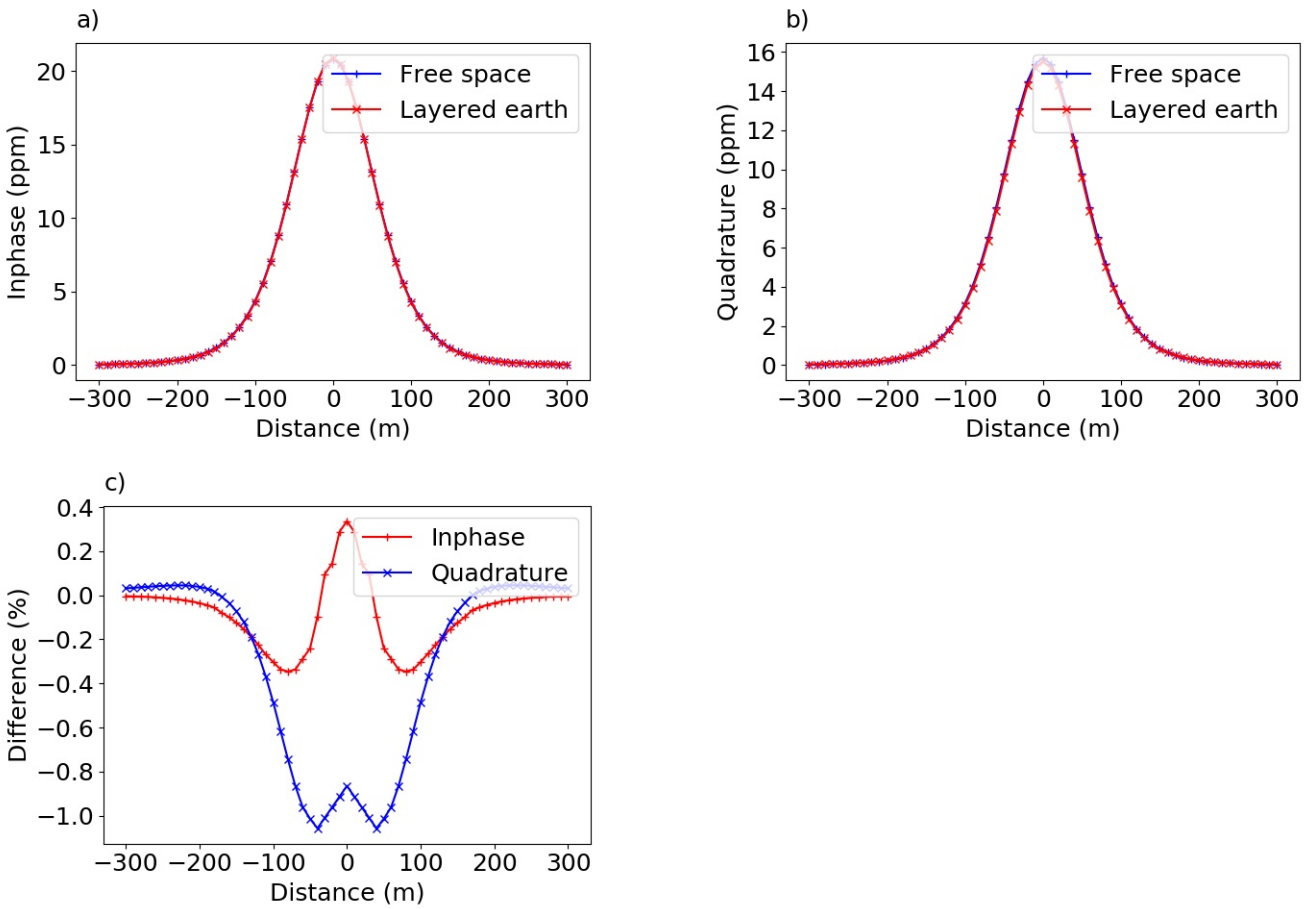

Figure 2: Sphere in free space and layered earth a) inphase, b) quadrature, and c) percent difference (relative to the maximum inphase) as a function of distance above a conductive sphere with its center located 100 m below the surface. This sphere has a radius of 50 m and conductivity of 1 S/m. The host rock conductivity is 0.1 mS/m. The transmitter loop altitude is 40 m and the frequency 1 kHz.

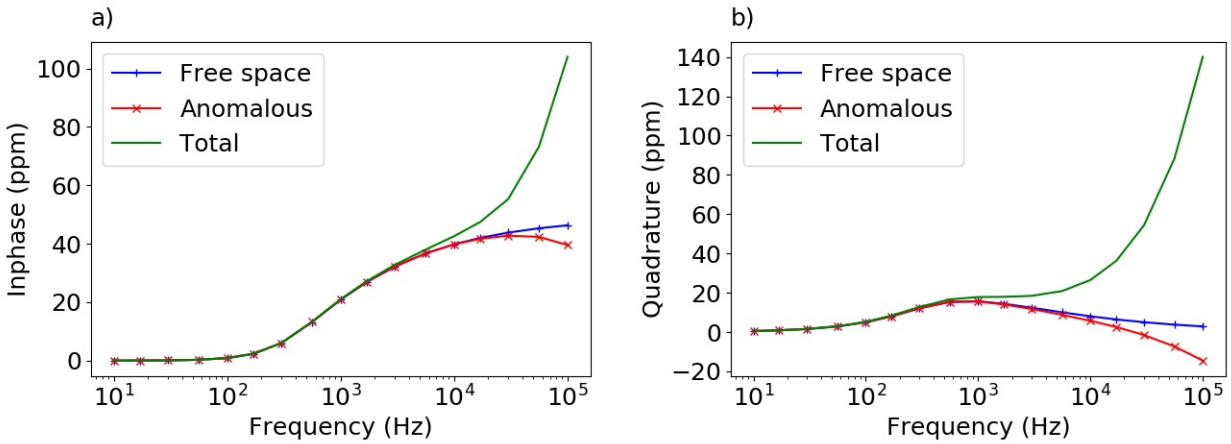

Figure 3: a) Inphase and b) quadrature field as a function of frequency above the top of a conductive sphere with its center located 100 m below the surface. This sphere has a radius of 50 m and conductivity of 1 S/m. The host rock conductivity is 0.1 mS/m. The transmitter loop altitude is 40 m. Shown are responses for the sphere in free-space ('Free space'), the sphere in a layered earth ('Total') and the latter minus the layered-earth response ('Anomalous').

distinct programs, Airbeo (Raiche et al. 2007, using Christensen 2010 Hankel filters and Raiche 1998 frequency to time domain transformation) and GA_AEM (Brodie 2016). The two programs have very similar responses excluding a very early time divergence.

The accuracy of the sphere solution improves with the number of terms included in the sum of spherical functions up to a certain limit. For a sphere in free space, Vallée (2015) demonstrated that the relative contribution of the higher degree terms decreases with the ratio of the sphere radius square over the transmitter-sphere receiver-sphere distance product. For AEM geometry, this ratio is small enough that only small degree terms need to be considered. We observed that the same rule applies for a sphere in a layered earth. For the models considered in this paper, contributions of functions higher than the sixth degree were smaller than 0.01 % of the sum. Consequently, the results presented in this paper were obtained with a sum of the first six degree functions.

Figure 5 presents a profile with the AirTEM system above a conducting sphere of 1 S/m and radius of 50 m, centred 100 m below the surface. The host rock conductivity is 1 mS/m. For host rock conductivities lower that 0.1 mS/m, the sphere response is identical to the response of

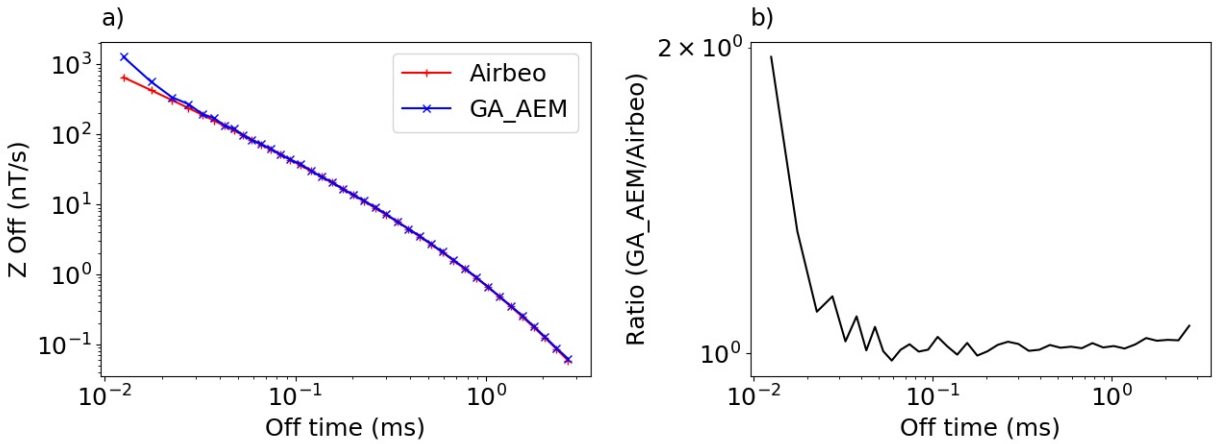

Figure 4: Time domain a) response and b) model ratio for Airbeo and GA_AEM programs, plotted as a function of off time above a conductive half-space of 1 mS/m. The AEM system is located 40 m above the ground.

a sphere in free space. The sphere anomaly peaks at early times around 500 nT/s and is a clear anomaly in the host rock background. "Total" response refers to the sum of the response of the host and the sphere, and "Anomalous" response refers to the response of the sphere only.

To illustrate the impact of the background conductivity on the response of the sphere, nomograms, representing a conductive sphere maximum response, have been computed for a sphere with identical parameters located in a conductive half-space or under a thin sheet of varying conductivity or conductance. The results are presented in Figures 6 and 7. In both models, the anomalous response is constant at low host conductivity, but decreases as the conductivity or conductance increases. For the two models, the anomalous field becomes negative before going to zero at high conductivities.

Figures 8 and 9 present results for a sphere located below a layer that is 50 m thick. In Figure 8, the conductivity of the overburden is 1 mS/m with a varying conductivity basement, while in Figure 9, the basement has a conductivity of 1 mS/m with a varying conductivity of the overburden. A similar pattern is observed compared to previous nomograms, with the response of the sphere decreasing as the conductivity and response of the host increases.

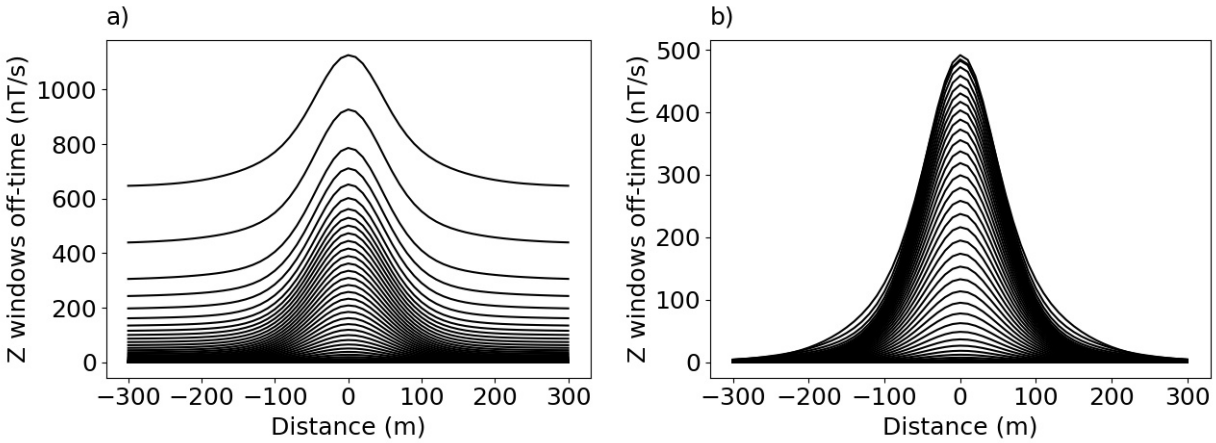

Figure 5: a) Total and b) anomalous field plotted as a function of distance above a conductive sphere with a radius of 50 m and conductivity of 1 S/m with its center located 100 m below surface. The host rock conductivity is 0.1 mS/m. The transmitter loop altitude is 40 m.

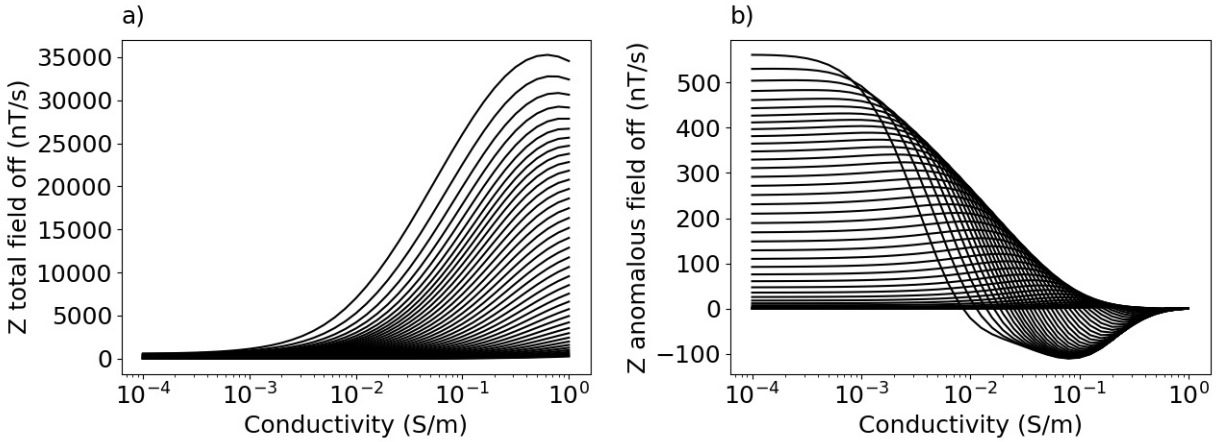

Figure 6: a) Total and b) anomalous maximum response as a function of half-space conductivity for a conductive sphere imbedded in a conductive half-space. The center of the sphere is located 100 m below the surface, with a radius of 50 m and conductivity of 1 S/m. Altitude of the transmitter loop is 40 m.

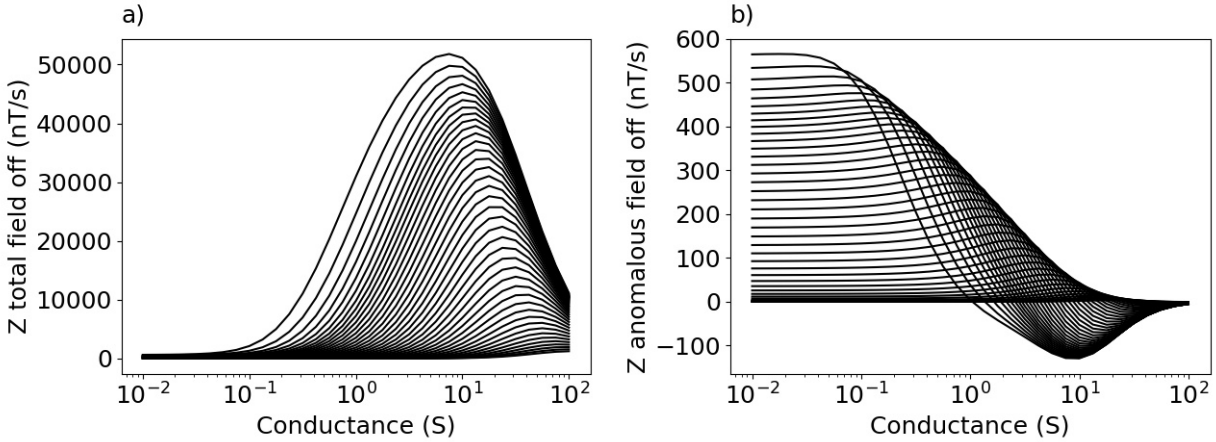

Figure 7: a) Total and b) anomalous maximum response as a function of thin-sheet conductance for a conductive sphere located below a conductive thin sheet located at surface. The host rock conductivity is 0.1 mS/m. The center of the sphere is located 100 m below the surface, with a radius 50 m and conductivity of 1 S/m. Altitude of the transmitter loop is 40 m.

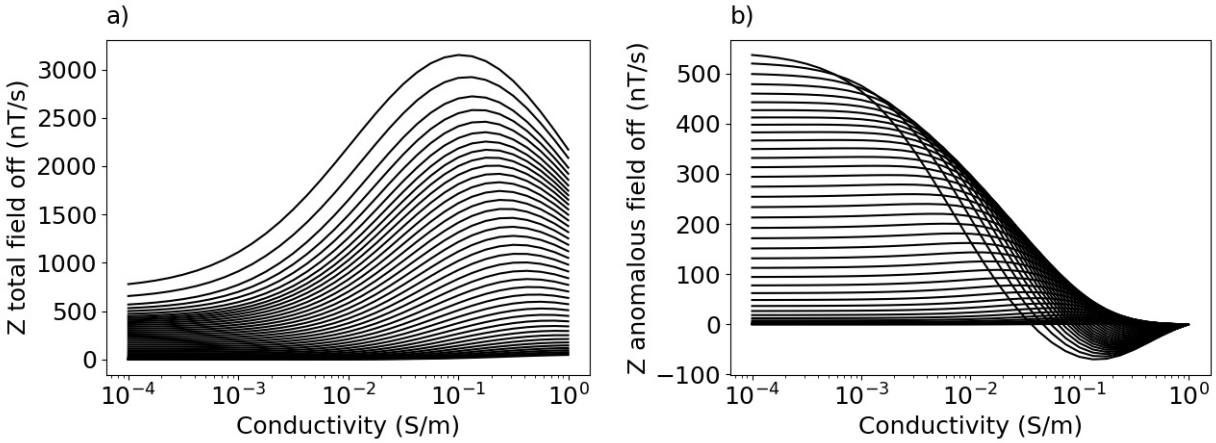

Figure 8: a) Total and b) anomalous maximum response as a function of basement conductivity for a conductive sphere located below an overburden of 50 m and 1 mS/m with varying basement conductivity. The center of the sphere is located 100 m below the surface, with a radius of 50 m and conductivity of 1 S/m. Altitude of the transmitter loop is 40 m.

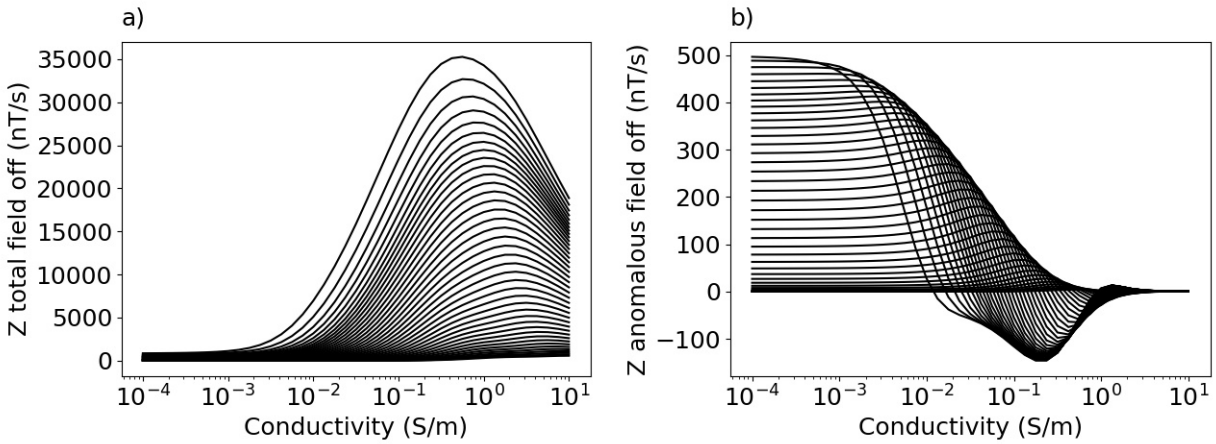

Figure 9: a) Total and b) anomalous maximum response as a function of overburden conductivity for a conductive sphere located below an overburden of 50 m and varying conductivity while the basement conductivity is 1 mS/m. The center of the sphere is located 100 m below the surface, with a radius of 50 m and conductivity of 1 S/m. Altitude of the transmitter loop is 40 m.

Field case

Modelling and inversion were tested across a significant anomaly observed with the AirTEM system at the Reid Mahaffy test site, Ontario, Canada (Witherly et al. 2004). This site is characterized by the presence of several conductors located in a resistive basement below a variable conductive overburden.

Figure 10 shows a color map of AirTEM 20 channel. We present interpretation for the large anomaly located on the North-West side of the survey area. Two lines spaced by 200 m located on the side and the center of the anomaly are also indicated. We used the technique of simulated annealing technique described by Yin and Hodges (2007) to estimate the electric parameters of the host rock and the overburden. From measurements located near the anomaly, we estimated a three-layer model with its parameters shown in Table 1, with a RMS error of around 3 %.

We then inverted for the parameters of the sphere model on the two selected lines. In doing so, we confirm the remarks of Chen and Raiche (1998) than SVD non-linear least square inversion is biased by the initial model. Figures 11 and 12 show the modeled and predicted responses for two different initial models: a shallow smaller sphere and a deeper larger sphere whose parameters

are given in Table 2. The smaller sphere presents the worst fit to the data, which translates into a higher RMS error. In each case, the iterations were stopped when the SVD inversion could not improve the solution.

An illustration of the model associated with the inversion of Figure 12 is shown in Figure 13. Notice that the AEM system is mostly sensitive to the first 200 m of the subsurface, and that below this depth the model is highly speculative. The most important conclusion drawn from this interpretation is that the conductor is both broad and with its top located slightly deeper than the overburden.

A sphere with the parameters of Table 2 is undetectable on Line 70 E if it is offset by 200 m. To fit the observations of Line 70 E, the sphere model must be either larger than expected or located directly under Line 70 E. Figure 14 shows the results from the inversion of a sphere located at 200 m from Line 70 E while Figure 15 shows inversion results for a sphere located directly under Line 70 E. Initial and inverted parameters for these inversions are given in Table 3. The best fit is achieved for a model located under Line 70 E. We interpret these results as limitations in the sphere model when more than one line observations are combined. We conclude that, in this situation, the sphere model can be used as an simple initial approximation for describing a more complex geological situation.

Layer	Conductivity (S/m)	Thickness (m)
1	0.0028	74.75
2	0.62	1.23
3	4.1e-5	

Table 1: Parameters of the layered earth.

Discussion

The models presented in this paper have several limitations. First, the host geology, in general, is more heterogeneous than an isotropic layered earth. Furthermore, the target, generally,

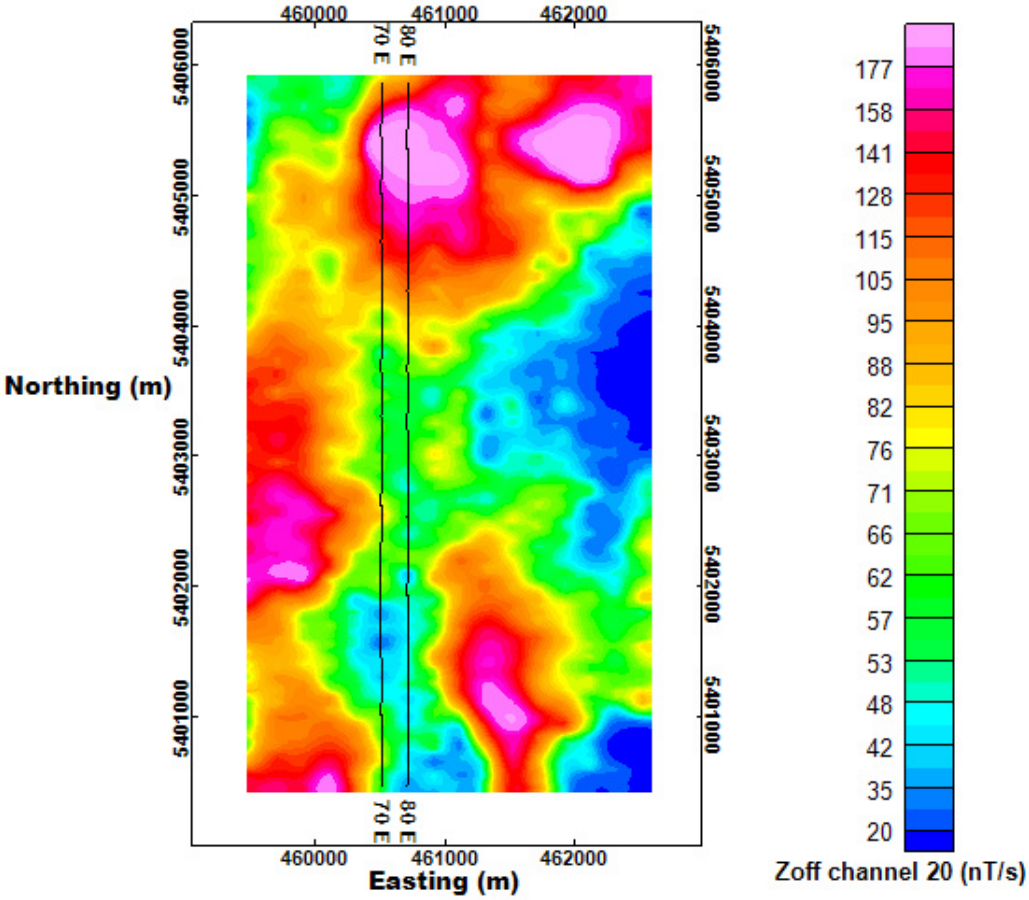

Figure 10: Color map of AirTEM channel 20 over Reid Mahaffy test site. Two lines crossing an important anomaly North of the area are indicated. Distance units are UTM 17N WGS 84.

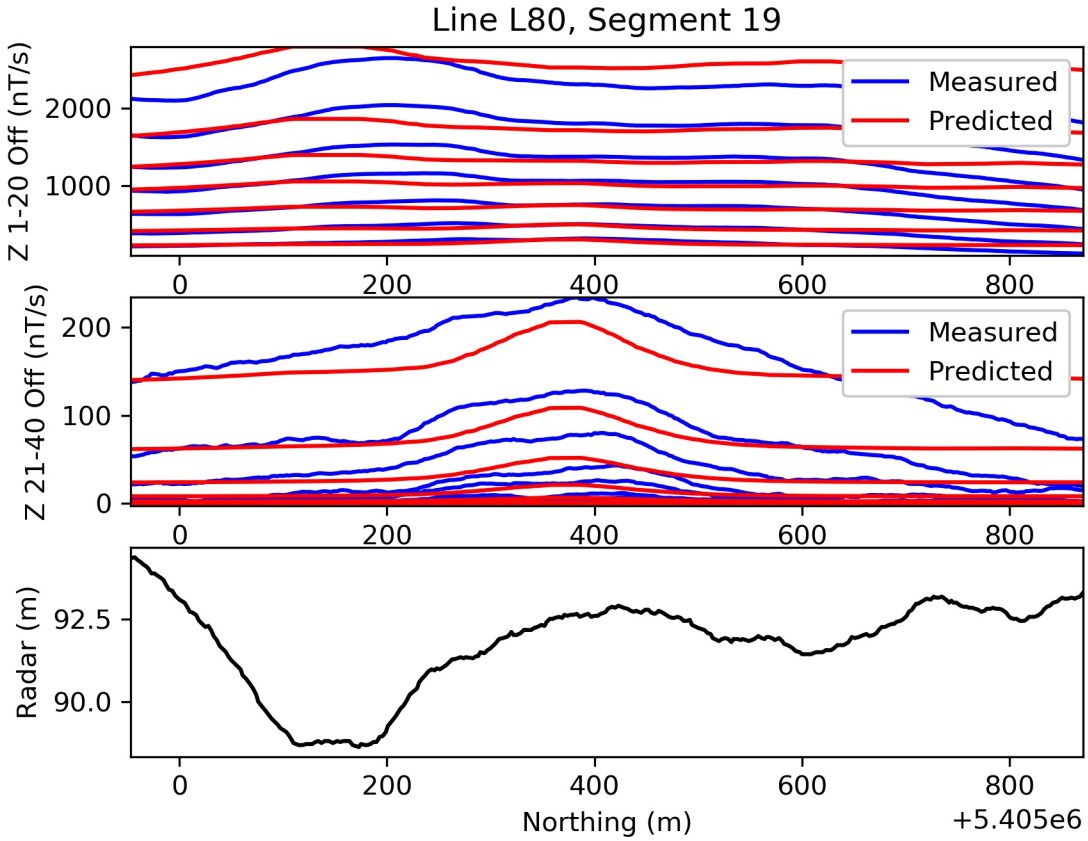

Figure 11: Measured (blue) and predicted (red) AEM response on Reid Mahaffy Line 80 E for a shallow small sphere. Initial and inverted parameters are given in Table 2. The radar is located on the aircraft 50 m above the towed AEM system. Only every third channel in presented.

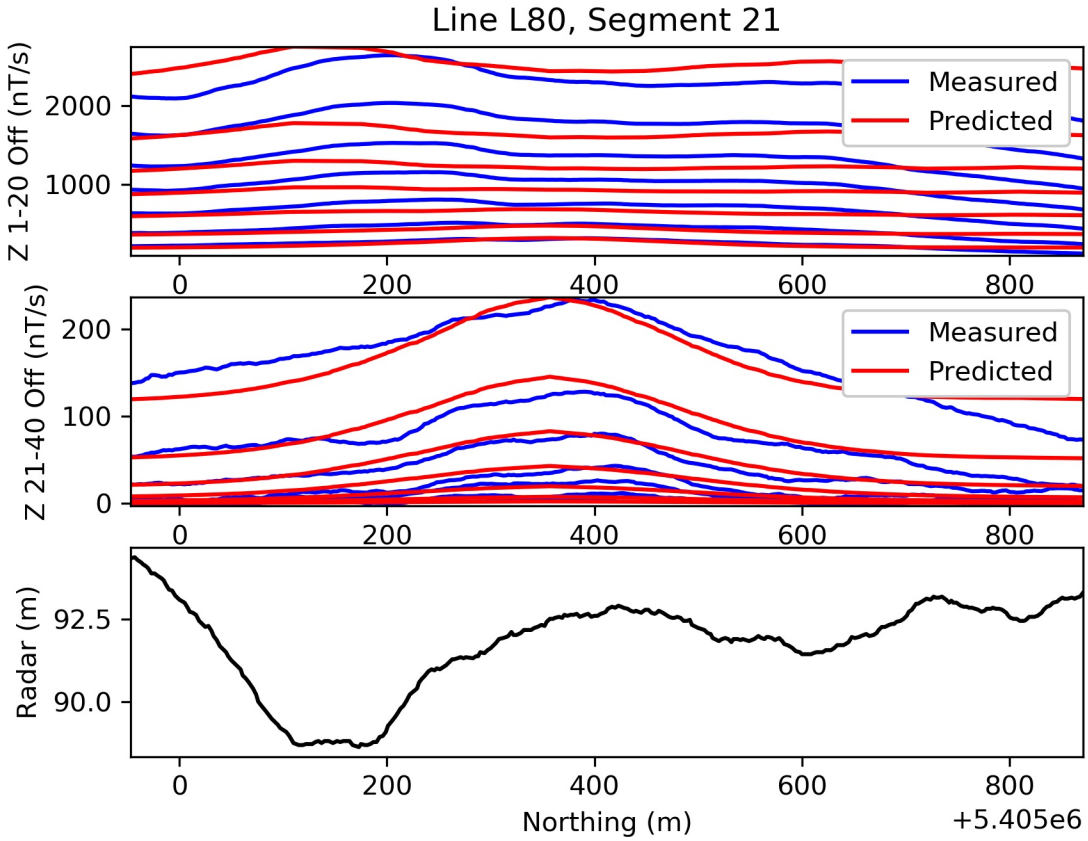

Figure 12: Measured (blue) and predicted (red) AEM response on Reid Mahaffy Line 80 E for a deep large sphere. Initial and inverted parameters are given in Table 2. The radar is located on the aircraft 50 m above the towed AEM system. Only every third channel in presented.

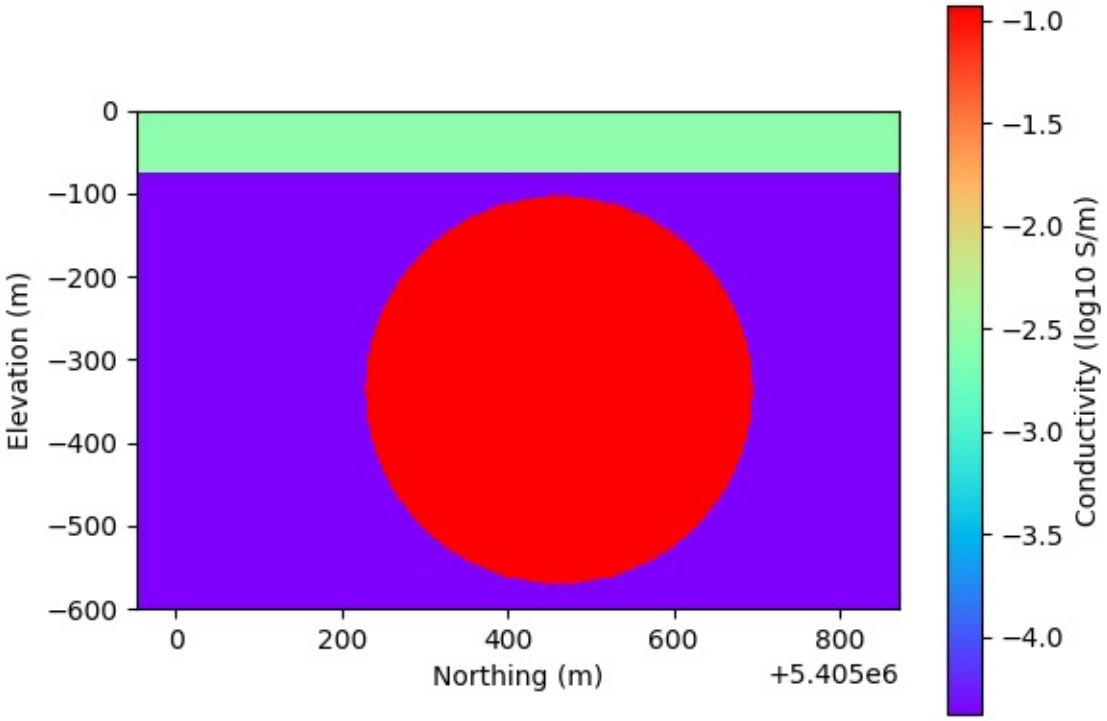

Figure 13: Illustration of the model of sphere in a layered earth where inverted data are shown on Figure 12 and parameters are given in Table 2.

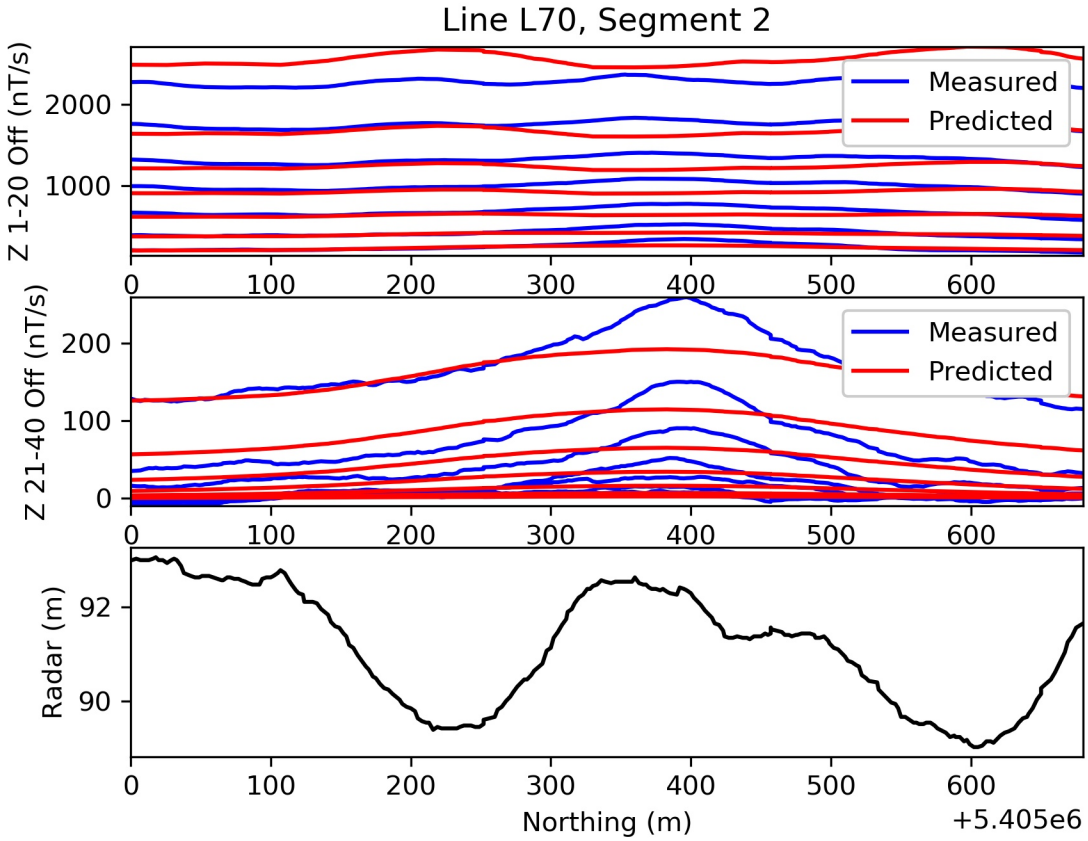

Figure 14: Measured (blue) and predicted (red) AEM response on Reid Mahaffy Line 70 E for a sphere offset 200 m. Initial and inverted parameters are given in Table 3. The radar is located on the aircraft 50 m above the towed AEM system. Only every third channel in presented.

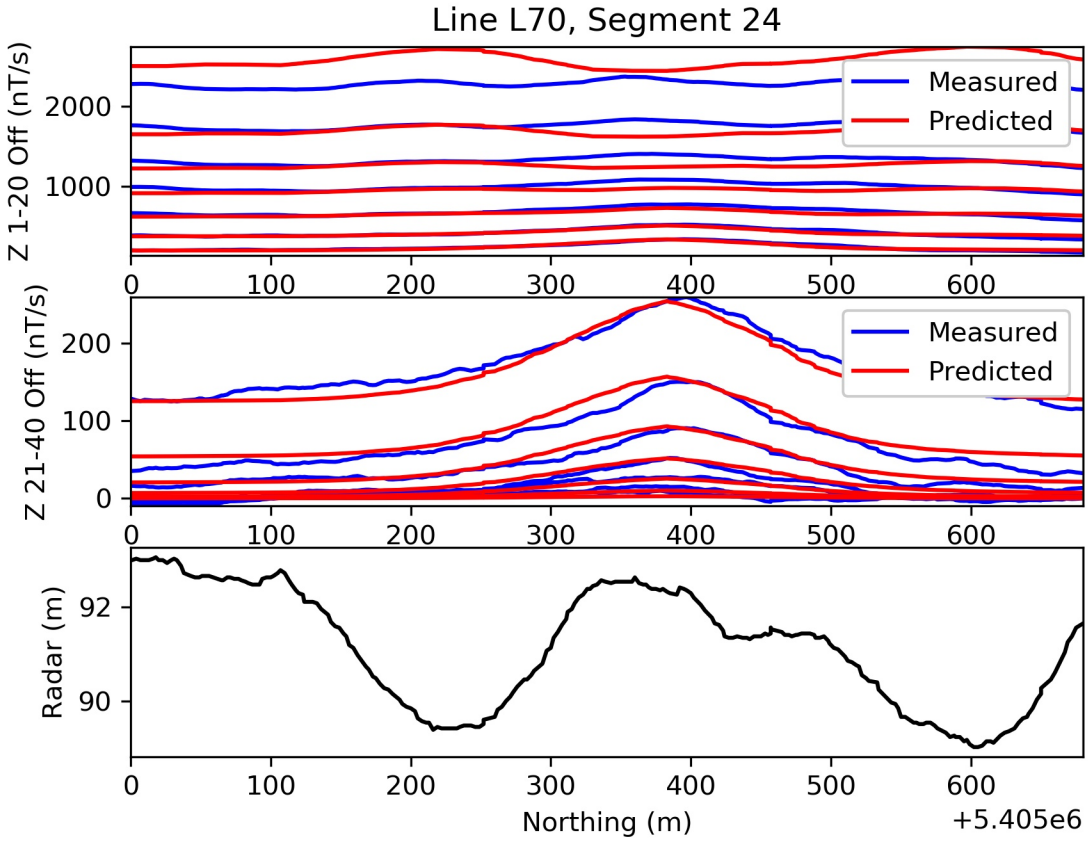

Figure 15: Measured (blue) and predicted (red) AEM response on Reid Mahaffy Line 70 E for a sphere located under the line. Initial and inverted parameters are given in Table 3. The radar is located on the aircraft 50 m above the towed AEM system. Only every third channel in presented.

Initial parameters			
	Conductivity (S/m)	1	1
	Radius (m)	50	300
	Center depth (m)	150	400
Inverted parameters			
	Conductivity (S/m)	1.1	0.11
	Radius (m)	49	232
	Center depth (m)	125	336
Number of iterations		4	11
RMS error (%)		27.8	15.8

Table 2: Initial and inverted parameters of the inverted models on Reid Mahaffy Line 80 E.

Initial parameters			
	Conductivity (S/m)	1	1
	Radius (m)	420	240
	Center depth (m)	500	-330
	Offset (m)	200	0
Inverted parameters			
	Conductivity (S/m)	0.1	0.9
	Radius (m)	302	81
	Center depth (m)	378	157
Number of iterations		4	9
RMS error (%)		16.7	10.0

Table 3: Initial and inverted parameters of the inverted models on Reid Mahaffy Line 70 E.

has a more complex geometry and conductivity distribution than those of a uniform sphere. For example, in the field case presented, the AEM system is only sensitive to the top of the conductor. Accordingly, the bottom information on the conductor is highly speculative, although the sphere model predicts an extension to 600 m depth. We also observed that line to line observations are explained by different models. Finally, the mathematical solution is based on a sum of spherical functions for which we noticed numerical instabilities when the degree is larger than ten.

Despite these limitations, this model has its values for the following reasons. In electromagnetic exploration, we often observe that several distinct models can fit the same data. Furthermore, AEM exploration geometry allows the use of a small number of terms of spherical functions to solve the sphere response. Consequently, applying a sphere in a layered earth model can be a first pass in the interpretation of AEM data sets, particularly in the situation where there is little

priori information. The sphere model, while oversimplified, may on occasion usefully fit locally anomalous data as shown here. The information gained by utilizing this model can help in the development and application of more elaborate models, particularly in an exploration projects since more information is gained as these projects progress. Additional benefits of this method include the potential validation of numerical models. In general, we should aim for an agreement near 1 %.

Conclusions

Based on the development of spherical and cylindrical functions, by using Cartesian coordinate derivatives as function expansions, an approach has been presented for relating spherical and cylindrical functions. This leads to a mathematical solution that determines the electromagnetic response of a conductive sphere in an isotropic layered earth, which is of interest in AEM exploration. An algorithm has been designed which subsequently led to the development of a computer program based on this design. This program allows the modelling and preliminary interpretation of AEM data. The algorithm may also have general applications in electromagnetic exploration geophysics for testing accuracy of new algorithms or software developed for modelling and inversion.

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A Appendix: Spherical function developments

Since there are various definitions of spherical functions, we review the definitions used in this paper. In general, we follow Gumerov and Duraiswami (2007) with the exception that the time variation is $e^{i\omega t}$ and the radial potentials are defined following Ward and Hohmann (1987). Electric and magnetic fields in spherical coordinates defined at locations $\mathbf{r} = \mathbf{f}(r, \theta, \phi)$ can be derived from two scalar potentials ψ_a and ψ_f as (Ward and Hohmann 1987, p. 281)

$$\mathbf{E} = -\nabla \times \mathbf{r}\psi_f + \frac{1}{\hat{y}}\nabla \times \nabla \times \mathbf{r}\psi_a \tag{A.1}$$

$$\mathbf{H} = \nabla \times \mathbf{r}\psi_a + \frac{1}{\hat{z}}\nabla \times \nabla \times \mathbf{r}\psi_f \tag{A.2}$$

where ψ_a and ψ_f are solutions of the Helmholtz equation. These solutions can be expressed in spherical coordinates as regular or inward $R_n^m(\mathbf{r})$ and singular or outward $S_n^m(\mathbf{r})$ functions. The regular functions are expressed as

$$R_n^m(\mathbf{r}) = j_n(kr)Y_n^m(\theta, \phi) \tag{A.3}$$

and the singular functions are

$$S_n^m(\mathbf{r}) = h_n(kr)Y_n^m(\theta, \phi) \tag{A.4}$$

where $j_n(kr)$ and $h_n(kr)$ are spherical Bessel and Hankel functions respectively where $Y_n^m(\theta, \phi)$ are the orthogonal spherical harmonics defined as

$$\begin{aligned} &Y_n^m(\theta, \phi) = (-1)^m \sqrt{\tfrac{2n+1}{4\pi}\tfrac{(n-|m|)!}{(n+|m|)!}} P_n^{|m|}(\cos\theta)e^{im\phi} \\ &P_n^{|m|}(\mu) = \tfrac{(-1)^{|m|}}{2^n n!}(1-\mu^2)^{|m|/2}\tfrac{d^{|m|+n}}{d\mu^{|m|+}}(\mu^2-1)^n \\ &n = 0, 1, 2, ...; m = -n, ..., n \end{aligned} \tag{A.5}$$

Gumerov and Duraiswami (2003) develop Cartesian derivative relations for the spherical functions. These relations are simpler for $\frac{\partial}{\partial x} \pm i\frac{\partial}{\partial y}$ and can be written as

$$\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right) T_n^m(\mathbf{r}) = k\left[b_{n+1}^{-m-1} T_{n+1}^{m+1}(\mathbf{r}) - b_n^m T_{n-1}^{m+1}(\mathbf{r})\right] \tag{A.6}$$

$$\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right) T_n^m(\mathbf{r}) = k\left[b_{n+1}^{m-1} T_{n+1}^{m-1}(\mathbf{r}) - b_n^{-m} T_{n-1}^{m-1}(\mathbf{r})\right] \tag{A.7}$$

$$\frac{\partial}{\partial z} T_n^m(\mathbf{r}) = k\left[a_{n-1}^m T_{n-1}^m(\mathbf{r}) - a_n^m T_{n+1}^m(\mathbf{r})\right] \tag{A.8}$$

and a_n^m and b_n^m are real coefficients which can be evaluated from the following expressions as

$$a_n^m = \begin{cases} \sqrt{\frac{(n+1+|m|)(n+1-|m|)}{(2n+1)(2n+3)}}, & |m| \leq n \\ 0, & |m| > n \end{cases} \tag{A.9}$$

$$b_n^m = \begin{cases} \sqrt{\frac{(n-m-1)(n-m)}{(2n-1)(2n+1)}}, & 0 \leq m \leq n \\ -\sqrt{\frac{(n-m-1)(n-m)}{(2n-1)(2n+1)}}, & -n \leq m < 0 \\ 0, & |m| > n \end{cases} \tag{A.10}$$

Developing derivative operators as matrix multiplications, we can write derivatives of spherical functions as (Gumerov and Duraiswami 2007)

$$\sum_{n=0}^{\infty}\sum_{m=-n}^{n} \hat{\psi}_n^m T_n^m(\mathbf{r}) = \sum_{n'=0}^{\infty}\sum_{m'=-n'}^{n'} \psi_{n'}^{m'} \frac{\partial}{\partial w} T_{n'}^{m'}(\mathbf{r}) = \sum_{n'=0}^{\infty}\sum_{m'=-n'}^{n'} \psi_{n'}^{m'} \frac{\partial}{\partial w} T_{n'}^{m'}(\mathbf{r}) \tag{A.11}$$

For $w = z$,

$$\sum_{n=0}^{\infty}\sum_{m=-n}^{n} \hat{\psi}_n^m T_n^m(\mathbf{r}) = \sum_{n'=0}^{\infty}\sum_{m'=-n'}^{n'} \psi_{n'}^{m'} \frac{\partial}{\partial z} T_{n'}^{m'}(\mathbf{r}) = \sum_{n'=0}^{\infty}\sum_{m'=-n'}^{n'} \psi_{n'}^{m'} k\left[a_{n'-1}^{m'} T_{n'-1}^{m'}(\mathbf{r}) - a_{n'}^{m'} T_{n'+1}^{m'}(\mathbf{r})\right] \tag{A.12}$$

The solution is

$$\hat{\psi}_n^m = k\delta_{m',m}\left(a_n^m \delta_{n',n+1}\psi_{n'}^{m'} - a_{n-1}^m \delta_{n',n-1}\psi_{n'}^{m'}\right) \tag{A.13}$$

where δ is the Kroneker delta. The general solution for $\hat{\psi}_n^m$ can be represented by the following sparse matrix

$$\hat{\psi}_n^m = \sum_{n'=0}^{\infty} \sum_{m'=-n'}^{n'} (D_w)_{nn'}^{mm'} \psi_{n'}^{m'} \tag{A.14}$$

where

$$\begin{aligned}
(D_{x+iy})_{nn'}^{mm'} &= k\delta_{m',m-1}(b_n^{-m}\delta_{n',n-1} - b_{n+1}^{m-1}\delta_{n',n+1}) \\
(D_{x-iy})_{nn'}^{mm'} &= k\delta_{m',m+1}(b_n^{m}\delta_{n',n-1} - b_{n+1}^{-m-1}\delta_{n',n+1}) \\
(D_z)_{nn'}^{mm'} &= k\delta_{mm'}(a_n^{m}\delta_{n',n+1} - a_{n'}^{m}\delta_{n',n-1})
\end{aligned} \tag{A.15}$$

Following Gumerov and Duraiswami 2007, these relations can be applied to various diffential operators and used to develop Cartesian field expansion from the spherical coefficients using equation (A.1) and (A.2). Expressed relative to the potentials ψ_a and ψ_f, these relations are:

$$(E_x)_n^m = \begin{array}{c} -\frac{i}{2}\left[c_n^{m-1}(\psi_f)_n^{m-1} + c_n^m(\psi_f)_n^{m+1}\right] \\ -\frac{k}{2\hat{y}} \left[\begin{array}{c} (n+2)b_{n+1}^{m-1}(\psi_a)_{n+1}^{m-1} + (n-1)b_n^{-m}(\psi_a)_{n-1}^{m-1} \\ +(n+2)b_{n+1}^{-m-1}(\psi_a)_{n+1}^{m+1} + (n-1)b_n^{m}(\psi_a)_{n-1}^{m+1} \end{array} \right] \end{array} \tag{A.16}$$

$$(E_y)_n^m = \begin{array}{c} \frac{1}{2}\left[-c_n^{m-1}(\psi_f)_n^{m-1} + c_n^m(\psi_f)_n^{m+1}\right] \\ +\frac{ik}{2\hat{y}} \left[\begin{array}{c} (n+2)b_{n+1}^{m-1}(\psi_a)_{n+1}^{m-1} + (n-1)b_n^{-m}(\psi_a)_{n-1}^{m-1} \\ -(n+2)b_{n+1}^{-m-1}(\psi_a)_{n+1}^{m+1} - (n-1)b_n^{m}(\psi_a)_{n-1}^{m+1} \end{array} \right] \end{array} \tag{A.17}$$

$$(E_z)_n^m = im(\psi_f)_n^m + \frac{k}{\hat{y}}\left[(n+2)a_n^m(\psi_a)_{n+1}^m + (n-1)a_{n-1}^m(\psi_a)_{n-1}^m\right] \tag{A.18}$$

$$(H_x)_n^m = \begin{array}{c} \frac{i}{2}\left[c_n^{m-1}(\psi_a)_n^{m-1} + c_n^m(\psi_a)_n^{m+1}\right] \\ -\frac{k}{2\hat{z}} \left[\begin{array}{c} (n+2)b_{n+1}^{m-1}(\psi_f)_{n+1}^{m-1} + (n-1)b_n^{-m}(\psi_f)_{n-1}^{m-1} \\ +(n+2)b_{n+1}^{-m-1}(\psi_f)_{n+1}^{m+1} + (n-1)b_n^{m}(\psi_f)_{n-1}^{m+1} \end{array} \right] \end{array} \tag{A.19}$$

$$(H_y)_n^m = \begin{array}{c} -\frac{1}{2}\left[c_n^{m-1}(\psi_a)_n^{m-1} + c_n^m(\psi_a)_n^{m+1}\right] \\ +\frac{ik}{2\hat{z}} \left[\begin{array}{c} (n+2)b_{n+1}^{m-1}(\psi_f)_{n+1}^{m-1} + (n-1)b_n^{-m}(\psi_f)_{n-1}^{m-1} \\ -(n+2)b_{n+1}^{-m-1}(\psi_f)_{n+1}^{m+1} - (n-1)b_n^{m}(\psi_f)_{n-1}^{m+1} \end{array} \right] \end{array} \tag{A.20}$$

$$(H_z)_n^m = -im(\psi_a)_n^m + \frac{k}{\hat{z}}\left[(n+2)a_n^m(\psi_f)_{n+1}^m + (n-1)a_{n-1}^m(\psi_f)_{n-1}^m\right] \tag{A.21}$$

where c_n^m is derived from the expansion development of $(\mathbf{r} \times \mathbf{w}) \cdot \nabla$, $\mathbf{w} = (x, y, z)$, and is defined as

$$c_n^m = \begin{cases} \sqrt{(n-m)(n+m+1)}, & 0 \leq m \leq n \\ -\sqrt{(n-m)(n+m+1)}, & -n \leq m < 0 \\ 0, & |m| > n \end{cases} \tag{A.22}$$

B Appendix: Cylindrical function developments

Spherical and cylindrical functions are related through the Sommerfeld (1909) integral which describes the field originating from a point source as a Hankel transform

$$\frac{e^{-ik}}{r} = \int_0^\infty \frac{\lambda}{u} e^{-u|z|} J_0(\lambda\rho) d\lambda \tag{B.1}$$

where k and u are defined below equations (4) and (5) respectively. This Hankel transform can be generalized as a cylindrical function by seeking solutions which are a summation of cylindrical functions

$$\sum_{m=-\infty}^{\infty} I_m(\rho, \phi, z) = \sum_{m=-\infty}^{\infty} \int_0^\infty \lambda J_{|m|}(\lambda\rho) e^{-u|z|} e^{im\phi} d\lambda \tag{B.2}$$

or

$$I_m(\rho, \phi, z) = \int_0^\infty \lambda G_m(\rho, \lambda, \phi, z) d\lambda \tag{B.3}$$

As with the spherical functions, we can develop derivatives expansions. From derivatives developed as

$$\begin{aligned} \tfrac{\partial}{\partial x} + i\tfrac{\partial}{\partial y} &= \tfrac{e^{i\phi}}{\rho}(\rho\tfrac{\partial}{\partial \rho} + i\tfrac{\partial}{\partial \phi}) \\ \tfrac{\partial}{\partial x} - i\tfrac{\partial}{\partial y} &= \tfrac{e^{-i\phi}}{\rho}(\rho\tfrac{\partial}{\partial \rho} - i\tfrac{\partial}{\partial \phi}) \end{aligned} \tag{B.4}$$

The derivatives of cylindrical functions are

$$
\begin{aligned}
(\tfrac{\partial}{\partial x} + i\tfrac{\partial}{\partial y})G_m(\rho,\lambda,\phi,z) &= \begin{cases} -\operatorname{sgn}(m)\lambda G_{m+1}(\rho,\lambda,\phi,z), & m \neq 0 \\ -\lambda G_1(\rho,\lambda,\phi,z), & m = 0 \end{cases} \\
(\tfrac{\partial}{\partial x} - i\tfrac{\partial}{\partial y})G_m(\rho,\lambda,\phi,z) &= \begin{cases} sgn(m)\lambda G_{m-1}(\rho,\lambda,\phi,z), & m \neq 0 \\ -\lambda G_{-1}(\rho,\lambda,\phi,z), & m = 0 \end{cases} \\
\tfrac{\partial}{\partial z}G_m(\rho,\lambda,\phi,z) &= -\operatorname{sgn}(z)uG_m(\rho,\lambda,\phi,z)
\end{aligned}
\tag{B.5}
$$

Cylindrical coefficient expansions can be developed from these relations. If we assume that

$$
\sum_{m=-\infty}^{\infty} \hat{\chi}_m G_m(\rho,\lambda,\phi,z) = \frac{\partial}{\partial w} \sum_{m'=-\infty}^{\infty} \chi_{m'} G_{m'}(\rho,\lambda,\phi,z) \tag{B.6}
$$

from equation (B.5) we derive a solution as

$$
\hat{\chi}_m = \sum_{m'=-\infty}^{\infty} (D_w)_{mm'} \chi_{m'} \tag{B.7}
$$

where

$$
\begin{aligned}
(D_{x+iy})_{mm'} &= \begin{cases} -\delta_{m,m'+1}\operatorname{sgn}(m')\lambda\chi_{m'} & m' \neq 0 \\ -\delta_{m,m'+1}\lambda\chi_{m'}, & m' = 0 \end{cases} \\
(D_{x-iy})_{mm'} &= \begin{cases} \delta_{m,m'-1}\operatorname{sgn}(m')\lambda\chi_{m'}, & m' \neq 0 \\ -\delta_{m,m'-1})\lambda\chi_{m'}, & m' = 0 \end{cases} \\
(D_z)_{mm'} &= -\delta_{m,m'}\operatorname{sgn}(z)u\chi_{m'}
\end{aligned}
\tag{B.8}
$$

Using these relations, electromagnetic fields can be represented as sums of cylindrical functions derived from potential coefficients associated with the different sources. Furthermore, using derivative relations, we can express singular spherical functions as cylindrical expansions where

$$
S_n^m(\mathbf{r}) = \sum_{m'=-\infty}^{\infty} \int_0^{\infty} \zeta_{nm'}^m G_{m'}(\rho,\lambda,\phi,z)\lambda d\lambda \tag{B.9}
$$

$\zeta_{nm'}^m$ is evaluated in the following way. First, the following recurrent relations can be derived for

equations (A.6) to (A.8).

$$S_n^m(\mathbf{r}) = \begin{cases} \frac{1}{kb_n^{-m}}(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y})S_{n-1}^{m-1}(\mathbf{r}), & m = n \\ \frac{1}{kb_n^{m}}(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y})S_{n-1}^{m+1}(\mathbf{r}), & m = -n \\ \frac{a_{n-2}^m}{a_{n-1}^m}S_{n-2}^m(\mathbf{r}) - \frac{1}{ka_{n-1}^m}\frac{\partial}{\partial z}S_{n-1}^m(\mathbf{r}), & -n < m < n \end{cases} \tag{B.10}$$

The recurrence start is derived from the Sommerfeld integral (B.1) using the definitions of the spherical singular functions (A.4) and (A.5)

$$\begin{aligned} S_0^0(\mathbf{r}) &= h_0(kr)Y_0^0(\theta,\phi) \\ &= -\frac{e^{-ikr}}{ikr\sqrt{4\pi}} \\ &= \frac{i}{k\sqrt{4\pi}}\int_0^\infty \frac{\lambda}{u}e^{-u|z|}J_0(\lambda\rho)d\lambda \\ &= \int_0^\infty \frac{i}{ku\sqrt{4\pi}}\lambda G_0(\rho,\lambda,z)d\lambda \end{aligned} \tag{B.11}$$

Then, the sparse matrix $\zeta_{nm'}^m$ can be developed from relations (B.10), (B.11), and (B.5).

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